Special instructions for invigilators: None.

Information for Candidates:

Numbers in brackets in the right margin (e.g. [5]) indicate maximum marks for each section of each question.

The following constants may be used:

electron charge: $e = 1.6 \times 10^{-19}$ C

permittivity of free space : $\varepsilon_o = 8.85 \times 10^{-12}$ F/m

relative permittivity of silicon : $\varepsilon_r = 12$

Planck's constant : $h = 6.63 \times 10^{-34}$ J s

Boltzmann's constant : $k = 1.38 \times 10^{-23}$ J/K

speed of light : $c = 3.0 \times 10^{8}$ m/s

The eigenvalue equations for TE modes in a symmetric slab waveguide of thickness d are

$\kappa = k_{1x} tan(k_{1x} d/2)$ and $\kappa = -k_{1x}\, cot(k_{1x} d/2)$

1. You should attempt all parts of this question. Short answers only are required; there is no need to re-state the questions in your answer book, but you should show any calculations you use to arrive at your answers, state any assumptions or approximations made, and give a brief (one or two lines) explanation where appropriate. All parts have equal value. [20]

a) Calculate the relationship between the refractive index and the relative permittivity of a dielectric material.

Solution: The speed of an electromagnetic wave in a material with refractive index n is:

$$u = \frac{c}{n}$$

where

$$c = \sqrt{\frac{1}{\varepsilon_0 \mu_0}}$$

And

$$u = \sqrt{\frac{1}{\varepsilon_r \varepsilon_0 \mu_r \mu_0}}$$

Dielectrics are usually non-magnetic, therefore $\mu_r = 1$, and hence:

$$n = \sqrt{\varepsilon_r}$$

b) If at some time instant a 2 km length of optical fibre contains 8000 bits of propagating information, what is the bit rate of this signal?

Solution:
Assuming that the optical fiber wave speed is c, the 2 km long information will arrive at the receiver within an interval of $t = \frac{L}{c} = \frac{2\ 10^3}{3 \cdot 10^8} = 6.67\ \mu s$

Therefore, the bitrate will be:

$$B = 8 \cdot \frac{10^3}{6.67 \cdot 10^{-6}} = 1.2\ Gbps$$

Note: In practice the speed of light in optic fibers is somewhat less. For $2 \cdot 10^8 \mathrm{m/s}$

$$B = 0.8\ Gbps$$

c) Visible light propagates through a sheet of uncoated window glass at approximately normal incidence. If the glass has a refractive index 1.49 in the visible range, estimate the fraction of optical power transmitted.

Solution:
The percentage of power that is reflected (i.e. lost) at each of the two interfaces will be:

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 = 3.8725\%$$

Hence, the fraction of optical power transmitted will be:

$$T_{total} = 1 - 2R = 92.3\%$$

d) A certain optical receiver detects $2 \cdot 10^{12}$ photons/s, at a nominal wavelength of 1.4 μm. What is the equivalent received optical power in dBm?

Solution:

The energy of each photon will be:

$$E_p = \frac{hc}{\lambda} = 1.42 \cdot 10^{-19} J$$

Therefore the power will be:
$P = 2 \cdot 10^{12} \cdot s^{-1} \cdot E_p = 284\ nW$
In dBm:

$$P_{dBm} = 10 \log\left(\frac{284nW}{1mW}\right) = -35.47\ dBm$$

e) Calculate the slope efficiency of a laser diode operating at λ_o = 1490 nm if the quantum efficiency is 0.8.

Solution:

$$\frac{d\Phi}{dI} = \frac{h\nu}{q}\eta$$

$$h\nu = h\frac{c}{\lambda_0} = 6.63 \cdot 10^{-34} Js \cdot \frac{3 \cdot 10^8 \frac{m}{s}}{1510 \cdot 10^{-9}\ m} = 0.133 \cdot 10^{-18} J$$

Hence:

$$\frac{d\Phi}{dI} = \frac{0.133 \cdot 10^{-18}}{1.6 \cdot 10^{-19}} \cdot 0.8 = 0.667\ \frac{J}{C} = 0.667\ W/A$$

f) Provide an expression for the noise power per unit bandwidth, for thermal noise in an optical detector.

Solution:
The square current per unit thermal noise bandwidth will be:

$$I^{*2} = \frac{4kT}{R}$$

Hence, the noise power per unit bandwidth in watt per hertz units will be:

$$\frac{P}{\Delta f} = I^{*2}R = 4kT$$

g) If thermal noise and shot noise are equal in a certain optical fibre link, and then the fibre length is decreased by 25%, which of these noise sources will now be greatest at the detector?

Solution:
Thermal noise originates from the receiver so it doesn't change.
Shot noise increases with signal intensity.
A shorter fibre will increase the received signal intensity and therefore the shot noise will then be greater that the thermal noise.

h) Explain briefly why silicon is not a favourable semiconductor for constructing optical sources.

Solution:
Silicon is an indirect bandgap seminconductor, meaning that a recombination requires a change of energy level and a change of momentum. Photons have very small momentum and therefore the generation of a phonon is difficult to happen. For this reason, recombination in indirect bandgap semiconductors is mostly non-radiating. This is what makes Si a poor optical source.

i) A silica optical fibre has a refractive index difference Δn of 0.03. Estimate the numerical aperture.

Solution:

$$NA = \sqrt{2\Delta n n_0}$$

Assuming a cladding refractive index of $n_0 = 1.5$:

$$NA = \sqrt{2 \cdot 0.003 \cdot 1.5} = \sqrt{0.09} = 0.3$$

j) Estimate the spectral width in nm of an LED operating at a nominal wavelength of 514 nm.

The spectral width is determined by the variation of recombination energy ΔE with at a temperature T is around 2kT.
The relation between ΔE and Δλ can be found from:

$$E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{E} \Leftrightarrow \frac{d\lambda}{dE} = -\frac{hc}{E^2} \Leftrightarrow \Delta\lambda = -\frac{\lambda^2}{hc}\Delta E = -\frac{\lambda^2}{hc}2kT = 11\ nm$$

2. A symmetric slab waveguide with dielectric and cladding refractive indices n_1 and n_2 respectively as shown in Fig. 2.1 has a core thickness d, and a numerical aperture $NA = 0.21$.

a) Find the value d/λ_o, where λ_o is the free space wavelength, for which the m=0 TE mode of this guide has an electric field distribution $E(x)$ such that $E(d/2) = E(0)/\sqrt{2}$. [10]

b) Does the guide that satisfies the condition in (a) support any additional TE modes, and if so how many? [5]

c) If the guide that satisfies the condition in (a) has a core thickness d = 5 μm, at what distance Δx beyond the boundary at $x = d/2$ does the electric field amplitude drop to ½$E(0)$? [5]

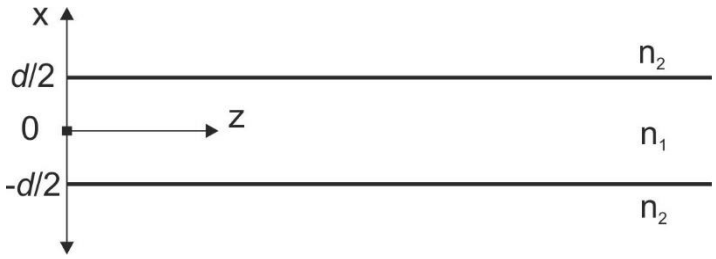


Figure 2.1

Solution:
a) The electric field will be:

$$E(x) = E(0)\cos(k_{1,x}x)$$

Hence:

$$E\left(\frac{d}{2}\right) = \frac{E(0)}{\sqrt{2}}$$

$$\cos\left(\frac{k_{1,x}d}{2}\right) = \frac{\sqrt{2}}{2} \Leftrightarrow k_{1,x}d = \frac{\pi}{2}$$

Also:

$$\kappa = k_{1,x}\tan\left(\frac{k_{1x}d}{2}\right) = \frac{\pi}{2d}$$

Finally:

$$\kappa^2 + k_{1,x}^2 = (n_1^2 - n_2^2)k_0^2 = NA^2k_0^2$$

But:

$$k_0 = 2\pi/\lambda_0$$

So, in total:

$$\frac{\pi^2}{2d^2} = \frac{NA^2 4\pi^2}{\lambda_0^2}$$

$$\frac{d}{\lambda_0} = \frac{1}{\sqrt{8}NA} = 1.68$$

b)

We saw in a) that:

$$NA^2k_0^2 = \frac{\pi^2}{2d^2}$$

To support a mode m, it must be:

$$NA^2k_0^2 \geq \left(\frac{m\pi}{d}\right)^2$$

This is not the case even for $\mathrm{m} = 1$. Therefore the waveguide does not support any other modes.

c)

For $x \geq \frac{d}{2}$

$$E(x) = E\left(\frac{d}{2}\right) \cdot e^{-\kappa\left(x-\frac{d}{2}\right)}$$

$$E(x) = \frac{E(0)}{\sqrt{2}} e^{-\kappa\left(x-\frac{d}{2}\right)}$$

We seek the Δx for which:

$$E\left(\frac{d}{2} + \Delta x\right) = \frac{E(0)}{2}$$

Hence:

$$\frac{E(0)}{\sqrt{2}} e^{-\kappa\Delta x} = \frac{E(0)}{2}$$

$$e^{-\kappa\Delta x} = \frac{\sqrt{2}}{2}$$

$$\kappa\Delta x = ln\sqrt{2} = 0.347$$

$$\Delta x = 0.347 \cdot 2 \cdot \frac{d}{\pi} = 1.1\ \mu m$$

3. A laser of peak output power 8 dBm, with nominal wavelength λ_o = 1550 nm and spectral width $\Delta\lambda$ = 0.4 nm, emits into a fibre with an attenuation of 0.2 dB/km and dispersion coefficient 0.75 ps/nm·km. It is modulated with an on-off keying format at a bit rate B. The receiver has quantum efficiency η = 0.8 and input resistance R = 20 kΩ.

a) Assume that the SNR is dominated by thermal noise in the receiver. Hence, derive an expression for the maximum bit rate as a function of fibre length, for an optical SNR of 12. Neglect dispersion, and state any other approximations or assumptions made. [5]

b) Assume that the SNR is dominated by shot noise in the receiver. Hence, derive an expression for the maximum bit rate as a function of fibre length, for an optical SNR of 12. Neglect dispersion, and state any other approximations or assumptions made. [5]

c) Derive an expression for the bit rate at which the dispersion time is equal to one quarter of the bit time $1/B$, as a function of fibre length. [5]

d) On a single graph of bit rate vs. fibre length, sketch the three relations derived in (a), (b) and (c) above, using appropriate scales and ranges for your two axes. Briefly discuss the significance of this graph for the operation of this optical link. [5]

Solution:

a) Thermal noise at the receiver:

$$I^{*2}_{th} = \frac{4kT}{R}$$

Then:

$$SNR^2 = \frac{I_{ph}^2}{I_{th}^{*\,2} \cdot \Delta f}$$

with $\Delta f = B/2$.

The signal at the receiver is:

$$I_{ph}^2 = R^2 \Phi_R^2 = \left(\frac{\eta e \lambda}{hc}\right)^2 \Phi_R^2 = \Phi_R^2$$

Hence:

$$\Phi_R^2 = SNR^2 I_{th}^{*\,2} \cdot \frac{B}{1.066 \cdot 2} = 6 \cdot 10^{-23} \cdot B$$

Power transport in the fibre:

$$8\ dBm - L \cdot \frac{0.2dB}{km} = 10log\Phi_R$$

$$16\ dBm - L \cdot \frac{0.4dB}{km} = 10log\Phi_R^2$$

$$16dB - 30dB - 0.4L = -230 + log6 + 10logB$$

$$logB = 21.6 - 0.04L$$

b) In the shot noise limited case:

$$I_{sh}^{*\,2} = 2eI_{ph}$$

Hence:

$$SNR^2 = \frac{I_{ph}^2}{I_{sh}^{*\,2} \cdot \Delta f} = \frac{I_{ph}}{eB} = \frac{R\Phi_R}{eB} = \frac{\eta\lambda}{Bhc}\Phi_R$$

$$B = \frac{\eta\lambda}{SNR^2 hc}\Phi_R = 4.33 \cdot 10^{16} \cdot \Phi_R$$

Again, the transmission in the fibre gives:

$$8dB - 30\,dB - 0.2 * L = 10log\Phi_R$$

Hence:

$$10logB = 10log4.33 + 160 + 10log\Phi_R$$

$$10logB = 138 + 6.36 - 0.2 \cdot L$$

$$logB = 14.4 - 0.02 \cdot L$$

c) The dispersion time at the receiver will be:

$$t_d = D \cdot L \cdot \lambda$$

This must be:

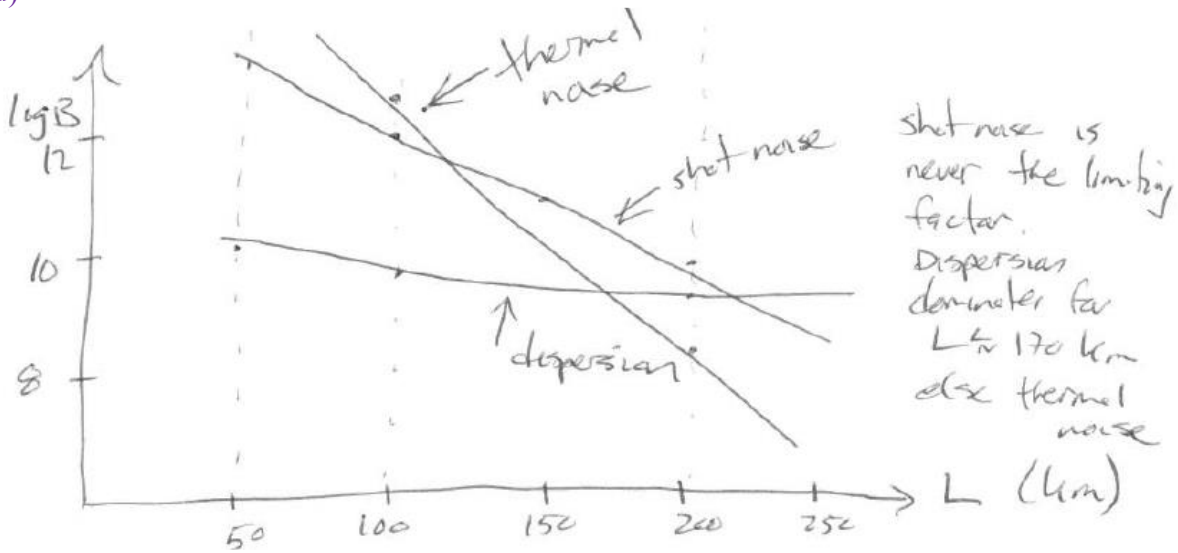
$$t_d = \frac{1}{4B}$$

Hence:

$$B = \frac{1}{4DL\Delta\lambda} = \frac{0.833\ km/ps}{L} = \frac{833}{L}GHz$$

$$logB = 11.9 - logL$$

d)



4. A given single mode fibre has a wavelength-dependent effective refactive index n' given by:

$$n' = n_g + \alpha(\lambda_o - \lambda_c)^2 \tag{4.1}$$

where λ_o is the free-space wavelength and λ_c is the centre operating range wavelength.

a) Calculate expressions for the phase delay τ_p, and the group delay τ_g, for a fibre length L. Note that the group velocity is given by $v_g = d\omega/d\beta$ where β is the propagation constant. [12]

b) Using your solution to (a), derive an expression for the dispersion coefficient D, using $D = \left|(d\tau_g/d\lambda_o)/L\right|$. Hence, show that in this case: $D = \left|\lambda_o(d^2n/d\lambda_o^2)/c\right|$.

Dispersion causes a frequency (or wavelength) dependent phase shift and therefore it can be modelled as an all-pass filter. Derive the expression for a filter function $H(\omega)$ equivalent to the dispersion effect of this fibre for a length L. [8]

Solution:

a)

The phase velocity is:

$$u_p = c/n' = \frac{c}{n_g + a(\lambda_0 - \lambda_c)^2}$$

The phase delay will then be:

$$\tau_p = \frac{L}{u_p} = \frac{L(n_g + a(\lambda_0 - \lambda_c)^2)}{c}$$

$$n' = \beta/k_0$$

The group velocity is:

$$u_g = \frac{d\omega}{dk} = \frac{d\omega}{d\beta}$$

$$\frac{\omega}{k_0} = c$$

$$\frac{2\pi}{\lambda_0} = \frac{\omega}{c}$$

$$\lambda_0 = 2\pi c/\omega$$

The group delay will be:

$$\tau_g = \frac{L}{u_g} = L\frac{d\beta}{d\omega}$$

$$\beta = n'k_0$$

$$\frac{d\beta}{d\omega} = \frac{d\beta}{d\lambda_0}\frac{d\lambda_0}{d\omega}$$

$$\frac{d\lambda_0}{d\omega} = -\frac{2\pi c}{\omega^2} = -\frac{\lambda_0}{\omega}$$

$$\frac{d\beta}{d\omega} = -\frac{\lambda_0}{\omega}\frac{d(n'k_0)}{d\lambda_0} = -\frac{\lambda_0}{\omega}\left[k_0\frac{dn'}{d\lambda_0} + n'\frac{dk_0}{d\lambda_0}\right]$$

$$\frac{d\beta}{d\omega} = -\frac{\lambda_0}{\omega}\left[k_0 2\alpha(\lambda_0 - \lambda_c) - (n_g + a(\lambda_0 - \lambda_c)^2)\frac{2\pi}{\lambda_0^2}\right]$$

$$\frac{d\beta}{d\omega} = -\frac{k_0}{\omega}\left[2\lambda_0\alpha(\lambda_0 - \lambda_c) - (n_g + a(\lambda_0 - \lambda_c)^2)\right]$$

$$\frac{d\beta}{d\omega} = -\frac{k_0}{\omega}\left[2\lambda_0^2\alpha - 2\lambda_0\alpha\lambda_c - n_g - a\lambda_0^2 + 2a\lambda_0\lambda_c - a{\lambda_c}^2\right]$$

$$\frac{d\beta}{d\omega} = -\frac{k_0}{\omega}\left[2\lambda_0^2\alpha - 2\lambda_0\alpha\lambda_c - n_g - a\lambda_0^2 + 2a\lambda_0\lambda_c - a{\lambda_c}^2\right]$$

$$\frac{d\beta}{d\omega} = -\frac{k_0}{\omega}\left[a(\lambda_0^2 - \lambda_c^2) - n_g\right]$$

Hence:

$$\tau_g = \frac{L}{c}a(\lambda_c^2 - \lambda_0^2) + n_g)$$

b)

$$\frac{d\tau_g}{d\lambda_0} = -\frac{2La\lambda_0}{c}$$

Hence:

$$D = \frac{2a\lambda_0}{c}$$

For:

$$n' = n_g + a(\lambda_0 - \lambda_c)^2$$

$$\frac{d^2n'}{d\lambda_0^2} = 2a$$

$$D = \left|\frac{\lambda_0}{c}\frac{d^2n'}{d\lambda_0^2}\right|$$

The filter should cause a phase shift of $\varphi = \tau_p\omega$, hence:

$$H(\omega) = e^{j\omega\tau_p} = e^{\frac{j\omega L}{c}n_g} \cdot e^{\frac{j\omega L}{c}a\left(\frac{2\pi c}{\omega} - \lambda_c\right)^2}$$

The first exponential term is the ω-independent delay, whereas the second term is the dispersive delay.

5. A silicon p-i-n photodiode has p and n doping levels respectively of $N_A = 2 \times 10^{21}$ m^{-3} and $N_D^+ = 10^{21}$ m^{-3}. The p-layer thickness is 0.6 µm. The attenuation coefficient in Si at the wavelength of interest is given by $\alpha = 0.2 \times 10^6$ m^{-1}.

a) Find the intrinsic layer thickness w_i such that 85% of photons are absorbed in the intrinsic layer (neglecting Fresnel reflection at the surface). [5]

b) Using w_i as calculated above, find the intrinsic layer doping level N_D^- such that the intrinsic region can be fully depleted by an applied voltage of 4.0 V. [5]

c) Using N_D^- and w_i as calculated above, find the applied bias voltage V such that the electric field amplitude in the intrinsic region varies by 20% (i.e. maximum value is 120% of minimum value). [5]

d) Discuss the main factors to be considered in optimising the electric field magnitude in a p-i-n photodiode. [5]

Solution:

a)
Neglecting surface (Fresnel) reflections:

$$I = I_0 \cdot e^{-ax}$$

$$\eta = \frac{E_{intrinsic}}{E_{total}} = \frac{\int_{wp}^{wp+wi} e^{-ax} dx}{\int_0^{+\infty} e^{-ax} dx} = e^{-awp} - e^{-a(wp+wi)}$$

$$1 - e^{-awi} = \eta e^{awp}$$

$$e^{-awi} = 1 - \eta e^{awp}$$

$$-awi = \ln\left(1 - \eta e^{awp}\right)$$

$$wi = -\frac{1}{a} \ln\left(1 - \eta e^{awp}\right)$$

$$wi = 15.9\ \mu m$$

b)
When the intrinsic area is just depleted, the field at the i-n junction should be zero.

Therefore:

$$E_{MAX} = -eN_D^- w_i / \varepsilon$$

And:

$$N_D^- w_i = N_A w_p$$

$$V = \frac{1}{2} w_p E_{MAX} + \frac{1}{2} w_i E_{MAX} = \frac{1}{2} E_{MAX} (w_p + w_i)$$

Hence:

$$V = \frac{1}{2\varepsilon} e N_D^- w_i (w_p + w_i) = \frac{1}{2\varepsilon} e N_D^- w_i \left(\frac{N_D^-}{N_A} w_i + w_i \right)$$

$$\frac{e w_i^2}{N_A} \cdot (N_D^-)^2 + e w_i^2 N_D^- = 2\varepsilon V$$

$$(N_D^-)^2 + N_A N_D^- - \frac{2\varepsilon V N_A}{e w_i^2} = 0$$

This gives:

$$N_D^- = 2.08 \cdot 10^{-19} m^{-3}$$

c)

In the intrinsic region:

$$\Delta E = \frac{q N_D^- w_i}{\varepsilon}$$

On the other hand:

$$E_1 = 1.2 E_2$$

And so:

$$E_1 = 6\Delta E$$

$$E_2 = 5\Delta E$$

$$E_{MAX} = -\frac{e}{\varepsilon} N_A w_p$$

$$w_p = \frac{\varepsilon E_1}{q N_A}$$

$$w_n = \frac{\varepsilon E_2}{q N_D^+}$$

The voltage needed for this condition will then be:

$$V = \int_0^{X2+wn} E dx = \frac{1}{2} w_p E_{MAX} + \frac{1}{2} w_i 1.83 E_{MAX} + \frac{1}{2} w_n 0.83 E_{MAX}$$

$$V = \frac{1}{2} \frac{\varepsilon}{q N_A} E_1^2 + \frac{1}{2} \frac{\varepsilon}{q N_D^-} \Delta E \cdot 11 \Delta E + \frac{1}{2} \frac{\varepsilon}{q N_D^+} E_2^2$$

$$V = \frac{\varepsilon}{2q} \cdot \left(\frac{E_1^2}{N_A} + 11 \frac{\Delta E^2}{N_D^-} + \frac{E_2^2}{N_D^+} \right)$$

$$V = \frac{\Delta E^2 \varepsilon}{2q} \cdot \left(\frac{36}{N_A} + \frac{11}{N_D^-} + \frac{25}{N_D^+}\right)$$

$$\Delta E = 498\ kV/m$$

$$V = 47.1\ V$$

d)

The field should be high enough to result in high carrier speed and therefore fast operation.
The field is limited by the junction breakdown effects.
Values between 10^6 and 10^7 V/m are generally sufficient for maximum drift velocity without breakdown.